

Social Acceleration: Ethical and Political Consequences of a Desynchronized High-Speed Society

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I. Social Acceleration in the Process of Modernization

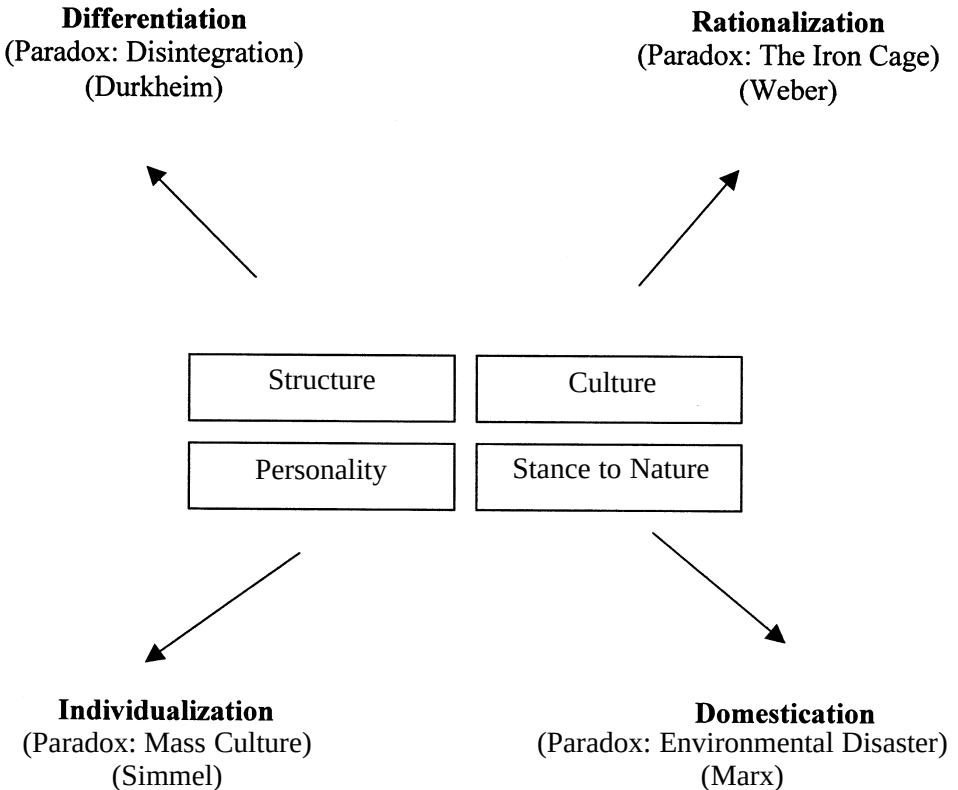
In 1999, James Gleick, exploring everyday life in contemporary American society, noted the “acceleration of just about everything”: love, life, speech, politics, work, TV, leisure, etc.¹ With this observation he certainly is not alone. In popular as well as scientific discourse about the current evolution of Western societies, *acceleration* figures as the single most striking and important feature.² But although there is a noticeable increase in the discourse about acceleration and the shortage of time in recent years, the feeling that history, culture, society, or even ‘time itself’ in some strange way *accelerates* is not new at all; it rather seems to be a constitutive trait of modernity as such. As historians like Reinhart Koselleck have persuasively argued, the general sense of a “speed-up” has accompanied modern society at least since the middle of the eighteenth century.³ And indeed, as many have observed and empirical evidence clearly suggests, the history of modernity seems to be characterized by a wide-ranging speed-up of all kinds of technological, economic, social, and cultural processes and by a picking up of the general pace of life. In terms of its structural and cultural impact on modern society, this change in the temporal structures and patterns of modernity appears to be just as pervasive as the impact of comparable processes of individualization or rationalization. Just as with the latter, it seems, social acceleration is not a steady process but evolves in waves (most often brought about by new technologies or forms of socio-economic organization), with each new wave meeting considerable resistance as well as partial reversals. Most often, a wave of acceleration is followed by a rise in the ‘discourse of acceleration,’ in which cries for deceleration in the name of human needs and values are voiced but eventually die down.⁴

However, contrary to the other constitutive features of the modernization process – *individualization*, *rationalization*, (functional and structural) *differentiation*, and the instrumental *domestication* of nature – which have all been the object of extensive analysis, the concept of *acceleration* still lacks a clear and workable definition and a systematic sociological analysis. Within systematic theories of modernity or modernization, *acceleration* is virtually absent, with the notable exception of Paul Virilio’s ‘dromological’ approach to history, which, alas, hardly amounts to a ‘theory.’ This surprising absence in the face of the

empirical and discursive omnipresence of processes of acceleration arguably is a reflection of the neglect of the temporal dimension and processual nature of society in twentieth-century sociological theory – a neglect noted by many authors, most famously perhaps by Anthony Giddens and Niklas Luhmann.⁵ In the history of sociology, modernization has mainly been analyzed from four different perspectives relating to culture, social structure, personality type, and the relation to nature.⁶ From these perspectives (prevalent, e.g., in the works of Weber, Durkheim, Simmel, and Marx, respectively), the process of modernization is identifiable as a process of *rationalization*, *differentiation*, *individualization*, or *instrumental domestication* respectively (Figure 1).

My claim here is that we cannot adequately understand the nature and character of modernity and the logic of its structural and cultural development unless we add the temporal perspective to our analysis. Of course, the temporal dimension runs across the four ‘material’ dimensions of society and cannot neatly be separated from them in phenomenological terms; there is no ‘social time’ independent of social structure, culture, etc. The dominant changes in the latter, i.e., individualization,

Figure 1: The Process of Modernization I



differentiation, rationalization, and domestication, are closely linked to the cardinal change in temporal patterns (acceleration), which appears both as their cause and their effect. In fact, it can be argued that many instances of the former are actually driven by the logic of acceleration. As I will briefly try to demonstrate in the closing section of this paper, the drive towards social acceleration in modern societies might in fact be so overriding that we might actually find phenomena of de-differentiation and de-individualization in cases where differentiation and individualization have become hindrances to social acceleration.

Interestingly, van der Loo and van Reijen state that for each of the four central processes of modernization there is a paradoxical flipside, which also has frequently been the focus of social analysis. For example, individualization goes hand in hand with the evolution of 'mass culture,' the overall result of rationalization could be imprisonment in a thoroughly irrational 'iron cage' (which, for example, is doomed to economic growth even when the only scarcity is the scarcity of scarcity), and the instrumental control and domination of nature could lead to a backlash in which manmade natural disasters wipe out our entire civilization. And sure enough, such a flipside is evident for social acceleration as well. Hence, no analysis of social acceleration is complete unless it takes into account those strange corresponding phenomena of social deceleration and slow-down that have become particularly visible towards the turn of the twenty-first century, with the rise of theories of 'hyper-acceleration,' 'turbo-capitalism,' and the 'digital speed-revolution' on the one hand, and conceptions of 'polar inertia,' the 'end of history,' the 'closing of the future,' and the sclerotic inescapability of the 'iron cage' on the other.⁷ From this latter perspective, all the apparent speed and transformation of society are only changes on the 'user surface,'⁸ beneath which processes of paralysis and sclerosis predominate.

But what is social acceleration? Does it refer to an acceleration of society itself, or does it only capture accelerating processes within (a static) society? In what sense can we talk of social acceleration in the singular, when all we see is a host of possibly unrelated processes of acceleration, e.g., in sports, fashion, video editing, transport, job-succession, as well as some phenomena of social deceleration? In the following I present an analytic framework that will allow, at least in principle, a theoretically thorough and empirically justifiable (or at least contestable) definition of what it could mean for a society to *accelerate* and of the ways in which Western societies can be understood as acceleration societies.

II. What Is Social Acceleration?

It is obvious that contrary to Gleick's observation of the 'acceleration of just about everything,' there is no single, universal pattern of acceleration that speeds up *everything*. To the contrary, many things *slow down*, like traffic in a traffic-jam, while others stubbornly resist all attempts to make them go faster, like the common cold. Nevertheless, there are certainly a great many social phenomena to

which the concept of acceleration can properly be applied. Athletes seem to be running and swimming faster and faster, computers compute at ever higher speeds, transport and communication need only a fraction the time they took a century ago, people appear to sleep less and less (some scientists found that the average sleeping time decreased by two hours since the nineteenth century and by 30 minutes since the 1970s⁹), and even our neighbors seem to move in and out of their flats more frequently.

But even if we can prove these changes are not accidental but follow a systematic pattern, is there anything these very different processes have in common such that they can be brought under the one concept of social acceleration? Not directly, I want to claim. Rather, when looking more closely at this range of phenomena, it becomes apparent that we can separate them into three analytically as well as empirically distinct categories. In the following I will first present these three categories of acceleration. In the next section, I will explore the connection between the different spheres of acceleration and the mechanisms or motors that lie behind them. In the fourth section, I will discuss some problems for the sociological analysis of 'acceleration-societies' that arise from the fact that we have to account for a range of social phenomena which remain constant or even decelerate. I will then discuss some of the most pressing and transformative political and ethical consequences of social acceleration before returning to the problem of the proper conceptualization of the process of modernization in the conclusion.

1) Technological Acceleration

The first, most obvious, and most measurable form of acceleration is the speeding up of intentional, *goal-directed* processes of transport, communication, and production that can be defined as *technological acceleration*. Although it is not always easy to measure the average speed of these processes, the general tendency in this realm is undeniable. Thus, the speed of communication is said to have increased by 10^7 , the speed of personal transport by 10^2 , and the speed of data processing by 10^6 .¹⁰

It is predominantly this aspect of acceleration that is at the center of Paul Virilio's 'dromology,' a narrative of historical acceleration which proceeds from the revolution in transport to that in transmission and finally to the 'transplantation' revolution dawning in the emergent possibilities of biotechnology.¹¹ The effects of technological acceleration on social reality are certainly tremendous. For example, the 'natural' (i.e. anthropological) priority of space over time in human perception (rooted in our sense organs and the effects of gravity, which allow for an immediate distinction of 'above' and 'below,' 'in front of' and 'behind,' but not of 'sooner' and 'later') seems to have been inverted: in the age of globalization and the u-topicality of the Internet, time is increasingly conceived as compressing or even annihilating space.¹² Space, it seems, virtually 'contracts' and loses its significance for orientation in the late modern world. Processes and

developments are no longer located and locations become ‘*non-lieux*,’ without history, identity, or relation.¹³

2) *Acceleration of Social Change*

Whereas phenomena of the first category can be described as acceleration processes *within* society, the phenomena of this second category could be classified as accelerations *of* society itself. When novelists, scientists, and journalists since the eighteenth century have observed the dynamization of Western culture, society, or history – and sometimes of time itself¹⁴ – they were not so much concerned with the spectacular technological advancements as with the (often simultaneous) accelerated processes of social change that rendered social constellations and structures as well as patterns of action and orientation unstable and ephemeral. The underlying idea is that rates of change themselves are changing. Thus, attitudes and values as well as fashions and lifestyles, social relations and obligations as well as groups, classes, or milieus, social languages as well as forms of practice and habits are said to change at ever increasing rates. This has led Arjun Appadurai to replace the symbolization of the social world as consisting of stable social aggregates which can be localized on *maps* with the idea of fluid, flickering *screens* representing cultural flows that only punctually crystallize into “ethno-, techno-, finan-, media- and ideoscapes.”¹⁵

However, empirically measuring (rates of) social change remains an unresolved challenge. There is little agreement in sociology as to what the relevant indicators of change are and when alterations or variations actually constitute a genuine or ‘basic’ social change.¹⁶ Here sociology might avail itself of approaches developed in social philosophy. German philosopher Hermann Lübbe claims that Western societies experience what he calls the “contraction of the present” (*Gegenwartsschrumpfung*) as a consequence of the accelerating rates of cultural and social innovation.¹⁷ His measure is as simple as it is instructive: for Lübbe, the *past* is defined as *that which no longer holds/is no longer valid* while the future denotes *that which does not yet hold/is not yet valid*. The present, then, is the time-span for which (to use an idea developed by Reinhart Koselleck) the horizons of experience and expectation coincide. Only within these time-spans of relative stability can we draw on past experiences to orient our actions and infer conclusions from the past with regard to the future. Only within these time-spans is there some certainty of orientation, evaluation, and expectation. In other words, *social acceleration is defined by an increase in the decay-rates of the reliability of experiences and expectations and by the contraction of the time-spans definable as the ‘present.’* Now, according to Lübbe, we can apply this measure of stability and change to social and cultural institutions and practices of all kinds: the present contracts in the political as well as the occupational, the technological as well as the aesthetic, the normative as well as the scientific or cognitive dimensions, i.e., in cultural as well as in structural respects.

But how could we verify this empirically? There seems to be fairly general agreement in the social sciences that the basic structures of society are those that organize the processes of production and reproduction. For Western societies since the early-modern period, these essentially include the family and the occupational system. And in fact, most studies of social change focus on exactly these domains, along with political institutions and technology. I will later turn to the question of how technological and social change, and hence technological acceleration and the acceleration of social change, are interrelated. For the moment I want to suggest that change in these two realms – family and work – has accelerated from an *inter-generational* pace in early-modern society to a *generational* pace in ‘classical modernity’ to an *intra-generational* pace in late modernity. Thus, the ideal-typical family structure in agrarian society tended to remain stable over the centuries, with generational turnover leaving the basic structure intact. In classical modernity, this structure was built to last for just a generation: it was organized around a couple and tended to disperse with the death of the couple. In late modernity, there is a growing tendency for family life-cycles to last less than an individual lifespan: increasing rates of divorce and remarriage are the most obvious evidence for this.¹⁸ Similarly, in the world of work, in premodern societies the father’s occupation was inherited by the son – again, potentially over many generations. In classical modernity, occupational structures tended to change with generations: sons (and daughters) were free to choose their own profession, but they generally chose only once, i.e., for a lifetime. In late modernity, occupations no longer extend over the whole of a work-life; jobs change at a higher rate than generations.¹⁹

If we try to formulate the argument more generally, the stability of social institutions and practices could serve as a yardstick for the acceleration (or deceleration) of social change. In the work of authors like Peter Wagner and Beck, Giddens, and Lash, theoretical as well as empirical support can be found for the thesis that institutional stability is generally on the decline in late-modern societies.²⁰ In a sense, the whole discourse about postmodernity and contingency hinges on this idea, although for now it is only meant to serve as a starting point for future empirical research.

3) *Acceleration of the Pace of Life*

Interestingly, there is a third type of acceleration in Western societies that is neither logically nor causally entailed by the first two, but rather seems, at least at first glance, paradoxical with respect to technological acceleration. This third process is the acceleration of the *pace of (social) life*, which has been postulated again and again in the process of modernity (e.g., by Simmel or, more recently, by Robert Levine).²¹ It is the central focus of much of the discussion about cultural acceleration and the alleged need for deceleration. Now, if we assume that ‘the pace of life’ – an admittedly fuzzy concept – refers to the speed and

compression of actions and experiences in everyday life, it is hard to see how it relates to technological acceleration. Since the latter describes the *decrease* of the time needed to carry out everyday processes and actions of production and reproduction, communication and transport, it should entail an *increase* in free time, which in turn would *slow down* the pace of life. Since technological acceleration means that less time is needed, time should become *abundant*. If, to the contrary, time becomes more and more *scarce*, this is a paradoxical effect that calls for a sociological explanation.²²

But first we must be able to measure the pace of life.²³ In my view, attempts to do so could follow a 'subjective' or an 'objective' approach, with the most promising route probably being a combination of the two. On the 'subjective' side, an *acceleration* of the speed of life (as against the speed of life itself) is likely to have effects on individuals' experience of time: it will cause people to consider time as scarce, to feel hurried and under time pressure and stress. Typically, people will feel that time goes by faster than before and they will complain that 'everything' goes too fast; they will worry that they might not be able to keep up with the pace of social life. Hence, the fact that this complaint has accompanied modernity ever since the eighteenth century does not prove that the speed of life was high all the time – in fact it does not help to determine 'the' speed of life at all – but it does hint at its continuous *acceleration*. As we might expect, recent studies indicate that in fact people in Western societies *do* feel under heavy time-pressure and they *do* complain about the scarcity of time. These feelings seem to have increased over recent decades,²⁴ making plausible the argument that the 'digital revolution' and the processes of globalization amount to yet another wave of social acceleration.²⁵

On the 'objective' side, an acceleration of the 'speed of life' can be measured in two ways. First, it should lead to a measurable contraction of the time spent on definable episodes or 'units' of action like eating, sleeping, going for a walk, playing, talking to one's family, etc., since 'acceleration' implies that we do *more* things in *less* time. This is a domain where time-use studies are of the highest importance. And in fact, some studies have found plenty of evidence for this: thus, for example, there appears to be a clear tendency to eat faster, sleep less, and communicate less with our families than our ancestors did.²⁶ Nevertheless, one needs to be very careful with such results: first, because the data for longitudinal time-use studies is extremely limited; second, because we always find counter-instances (e.g. the time fathers spend with their children in at least some sections of Western societies is clearly increasing) without being able to adequately determine the significance of these findings; and third, because it is frequently unclear what *drives* the measured accelerations (e.g., that people on average sleep less today than previous generations might simply be attributable to the fact that they grow older and don't work as hard physically). The second way to 'objectively' explore the acceleration of the pace of life consists in measuring the social tendency to 'compress' actions and experiences, i.e., to do and experience more

within a given period of time by reducing the pauses and intervals and/or by doing more things simultaneously, like cooking, watching TV, and making a phone call at the same time.²⁷

III. What Drives Social Acceleration?

When looking for the social forces that drive the wheels of acceleration, it becomes necessary to rethink the connection between the three 'spheres' of acceleration discussed so far. The major problem here lies in the paradox of the simultaneity of technological acceleration (1) and the increasing scarcity of time (3). If free time decreases in spite of technological acceleration, the only possible explanation is that the quantity of activity itself has changed, or more precisely, has risen faster than the corresponding technological rate of acceleration. Thus, free time is produced when the technological acceleration rate is above the rate of growth, where 'growth' refers to all kinds of time-consuming actions and processes. Conversely, time becomes scarce when growth rates are higher than the acceleration rates. For example, when the speed of transport doubles, half of the time previously used for transport is available as 'free time.' However, if speed doubles while the distance we need to cross quadruples, we need twice as much time as we did before: time becomes scarce. The same holds for processes of production, communication, etc. It is important to note that growth and acceleration are neither logically nor causally interconnected, since only the acceleration of constant processes logically entails a corresponding augmentation, whereas processes of transport, communication, or production are not necessarily constant. *Hence, we should apply the term 'acceleration society' to a society if, and only if, technological acceleration and the growing scarcity of time (i.e., an acceleration of the 'pace of life') occur simultaneously, i.e., if growth rates outgrow acceleration rates.*

Now this, interestingly, is one way in which the acceleration of the pace of life and technological acceleration are interconnected: technological acceleration can be seen as a social *answer* to the problem of scarce time, i.e., to the acceleration of the 'pace of life.' When we examine the causal relations between the three spheres of social acceleration, a surprising feedback loop is revealed: technological acceleration, which is frequently connected to the introduction of new technologies (like the steam engine, the railway, the automobile, the telegraph, the computer, the Internet), almost inevitably brings about a whole range of changes in social practices, communication structures, and corresponding forms of life. For example, the Internet has not only increased the speed of communicative exchange and the 'virtualization' of economic and productive processes; it also establishes new occupational, economic, and communicative structures, opening up new patterns of social interaction and even new forms of social identity.²⁸ Hence, it is easy to see how and why technological acceleration is prone to go hand in hand with the acceleration of change in the form of changing social structures and patterns,

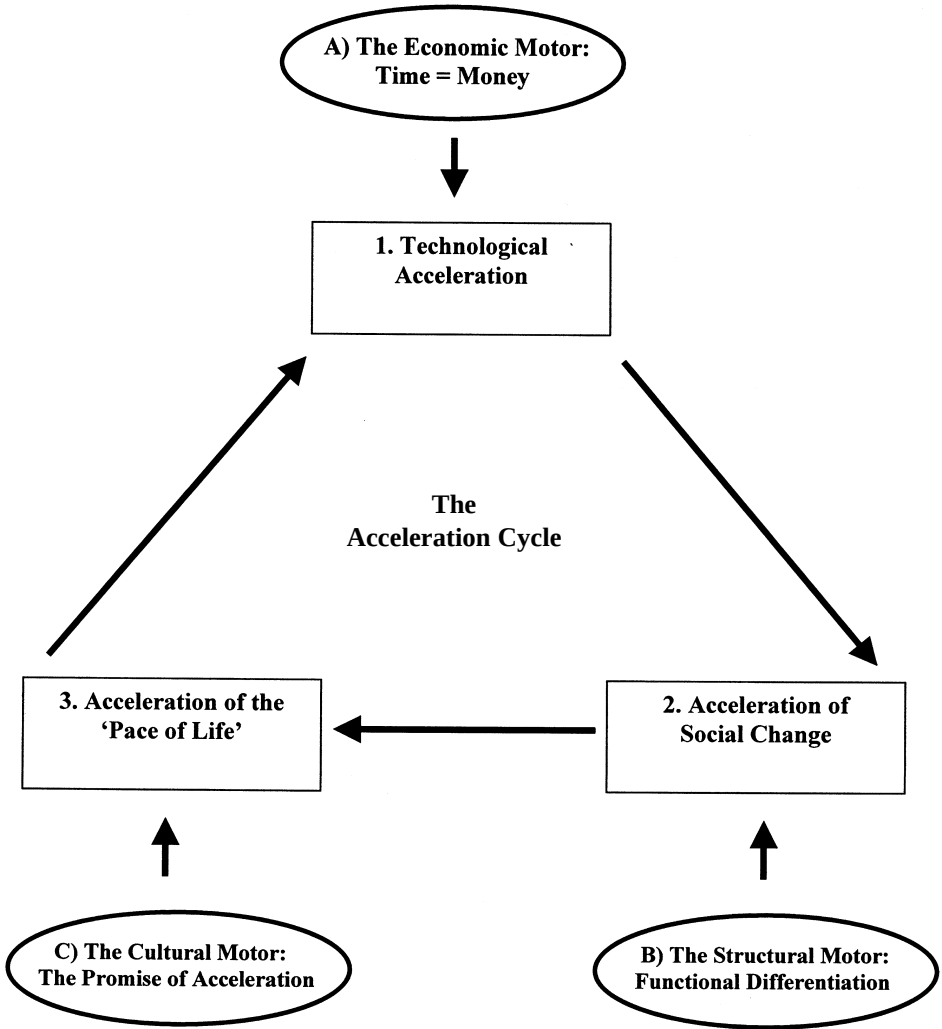
orientations, and evaluations of action. Furthermore, if the acceleration of social change entails a 'contraction of the present' in the sense discussed above, this naturally leads to an acceleration of 'the pace of life.' The explanation for this is to be found in a phenomenon that is well known from the realm of capitalist production and might be called the 'slippery slope' phenomenon: the capitalist cannot pause and rest, stop the race and secure his position, since he either goes up or goes down; there is no point of equilibrium since *standing still* is equivalent to *falling behind*, as Marx as well as Weber pointed out. Similarly, in a society with accelerated rates of social change in all spheres of life, individuals always feel that they stand on 'slippery slopes': taking a prolonged break means becoming old-fashioned, out-dated, anachronistic in one's experience and knowledge, in one's equipment and clothing as well as in one's orientations and even in one's language.²⁹ Thus, people feel pressed to keep up with the speed of change they experience in their social and technological world in order to avoid the loss of potentially valuable options and connections (*Anschlußmöglichkeiten*). This problem is aggravated by the fact that in a world of incessant change it gets increasingly difficult to tell *which* options will eventually turn out to be valuable. Hence, accelerated social change will in turn lead to an acceleration of the 'pace of life.' And finally, as we saw at the outset, new forms of technological acceleration will be called for to speed up the processes of productive and everyday life. Thus the 'acceleration cycle' is a closed, self-propelling process (Figure 2).

Yet the acceleration cycle by itself is not sufficient to explain the inherent dynamics of Western societies, or to understand its origins and the specific ways in which the logic and dynamic of speed and growth are interwoven. When looking for the driving forces of acceleration beyond the feedback cycle itself, one finds that there are three (analytically independent) primary factors that can be identified as the external 'key-accelerators' behind the three dimensions of social acceleration. In each of them, the logics of growth and speed are connected in a particular way characteristic of one of the dimensions of social acceleration.

1) *The Economic Motor*

The most obvious source of social acceleration in Western societies is, of course, capitalism. Within a capitalist economy, labor time figures as a crucial factor of production such that saving time is equivalent to making (relative) profit, as expressed in Benjamin Franklin's famous equation of time and money. Also, 'time leads' over competitors in the introduction of new technologies or products is a key element of market competition because it allows for crucial 'extra-profits' before the competitors catch up. Finally, the accelerated reproduction of invested capital is crucial with respect to what Marx called the 'moral consumption' of technology and to the credit system. As a consequence, the circle of production, distribution, and consumption constantly accelerates. This certainly explains the restless competition for *technological* acceleration in capitalist societies. In short, the

Figure 2: Motors of Acceleration



functioning of the capitalist system rests on the accelerating circulation of goods and capital in a growth-oriented society. Thus the logic of capitalism connects *growth* with *acceleration* in the need to increase production (growth) as well productivity (which can be defined in terms of time as *output per unit time*).

It is therefore not surprising that many authors concerned with the problem of social acceleration have attributed not only technological but *all* forms of acceleration to capitalism.³⁰ However, this assumption of enforced capitalist acceleration by itself seems insufficient to explain a whole range of acceleration

phenomena in dimensions (2) and (3), some of which reveal that processes of acceleration are by no means always or even usually enforced by competition, but frequently awaited with eudaimonistic or even eschatological undertones.³¹ This is where the *cultural motor* of acceleration comes in, a motor which seems indispensable for the explanation of the very success of capitalist forms of production.

2) The Cultural Motor

The acceleration of social change in Western societies is indissolubly linked with the dominant cultural ideals of modernity. These have gradually shifted the balance between tradition and innovation towards the priority of change such that 'real life,' as Friedrich Ancillon observed in 1828, is to be sought in *change for the sake of change*.³² Now, without denying that the evolution of industrial and capitalist forms of production and the accompanying social practices played a key role in the institutionalization of this idea, it is important to see that its roots reach further back. The ideal formulated by Ancillon is the consequence of a conception of life in which the *good life* is the *fulfilled life*, i.e., a life that is rich in experiences and developed capacities. This dominant modern cultural ideal evolved in the secularization of time and of conceptions of human happiness, analyzed at length by Hans Blumenberg and more recently by Marianne Gronemeyer and Gerhard Schulze.³³ The idea of the *fulfilled life* no longer supposes a 'higher life' waiting for us after death, but rather consists in realizing as many options as possible from the vast possibilities the world has to offer. *To taste life in all its heights and depths and in its full complexity* becomes a central aspiration of modern man.³⁴ But, as it turns out, the world always seems to have more to offer than can be experienced in a single lifetime. The options on offer always outgrow those realizable in an individual's life, or, in Blumenberg's terms, the perceived time of the world (*Weltzeit*) and the time of an individual life (*Lebenszeit*) dramatically diverge. *Acceleration of the pace of life* appears to be a natural solution to this problem: if we live 'twice as fast,' if we take only half the time to realize an action, goal, or experience, we can double what we can do within our lifetime. Our 'efficacy,' the proportion of *realized* options to potentially *realizable* options, doubles. It follows that in this cultural logic, too, the dynamics of growth and acceleration are intricately interwoven.

Now, on this cultural logic, if we keep increasing the speed of life, we could eventually live a multiplicity of lives within a single lifetime by taking up all the options that would define them. Acceleration serves as a strategy to erase the difference between the time of the world and the time of our life. *The eudaimonistic promise of modern acceleration thus appears to be a functional equivalent to religious ideas of eternity or 'eternal life,' and the acceleration of 'the pace of life' represents the modern answer to the problem of finitude and death.*

However, due to the self-propelling dynamic of the 'acceleration cycle,' the promise of acceleration never is fulfilled, for the very same techniques, methods,

and inventions that allow for an accelerated realization of options simultaneously increase the number of options (of 'world-time' or 'world-resources,' so to speak) at an exponential rate. For example, the Internet not only speeds up information and communication, it also opens up wholly new domains of exchange, service, communication, and entertainment. Hence, whenever we surf the net, we *could potentially* surf hundreds and thousands of other sites that might even better serve our purposes. The same holds true for cable TV: whereas 30 years ago we only missed two or three other programs by watching one channel, we now miss hundreds.³⁵ This, of course, has created the cultural phenomenon of 'zapping.' As a consequence, our share of the world, the proportion of *realized world options* to *potentially realizable ones*, decreases (contrary to the original promise of acceleration) no matter how much we increase the 'pace of life.' And this is the *cultural* explanation for the paradoxical phenomenon of simultaneous technological acceleration and increasing time scarcity.

3) The Structural Motor

Apart from the economic and the cultural explanations for the dynamics of modern Western acceleration, some sociologists have identified a third external engine in the *social structure* of modern society. According to this view, which is advocated predominantly in the context of Niklas Luhmann's systems theory, social change is accelerated by modern society's basic structural principle of *functional differentiation*. In a society that is not primarily segregated in hierarchical classes but rather structured along the lines of functional 'systems,' like politics, science, art, the economy, law, etc., complexity increases immensely. As a result, the future opens up to almost unlimited contingency and society experiences time in the form of perpetual change and acceleration.³⁶ Now, increasing complexity and contingency create an abundance of options and possibilities. Since these cannot be handled simultaneously, Luhmann argues that complexity in modern societies is 'temporalized' in order to enable the sequential processing of a higher number of options and relations than could be processed simultaneously. The ensuing needs for synchronization and selection of increasing (future) options can in turn only be satisfied if the processing itself is accelerated. Thus, we find a surprising structural duplication or 'reflection' of the cultural dilemma outlined in the preceding paragraph (or vice versa). Here, too, we find a variant of the internal dialectics of growth and acceleration that is characteristic of modern societies, here as a driving motor for (structural) *social change* (Figure 2).

IV. The Form and Relevance of Social Deceleration

Now, even if we find convincing evidence for acceleration in all three spheres defined above, it is crucial not to be drawn into a logic of subsumption where every social process or phenomenon is seen as determined by the dynamics of

acceleration. Hence, before we can adequately determine the sense in which we can speak of *the acceleration of Western societies*, we need to understand the status, function, and structure of those phenomena that escape dynamization or even represent forms of slow-down and deceleration. Analytically, we can distinguish five different forms of deceleration and inertia, which cut across the spheres of acceleration identified so far.

- 1) First, there are *natural* and *anthropological speed limits*. Some things cannot be accelerated in principle. Among these are most physical processes, like the speed of perception and processing in our brains and bodies, or the time it takes for most natural resources to reproduce.
- 2) Furthermore, there are territorial as well as social and cultural ‘niches’ that have not yet been touched by the dynamics of modernization and acceleration. They have simply been (totally or partially) exempted from acceleration processes, although they are accessible to them in principle. In such contexts, time seems to be ‘standing still,’ as the saying goes, e.g., forgotten islands in the sea, socially excluded groups or religious sects such as the Amish, or traditional forms of social practice (like producing whiskey in the famous Jack Daniels commercial). Arguably, these ‘*oases of deceleration*’ come under increasing pressure in late modernity unless they are *deliberately protected* against acceleration and thus fall under category (4).
- 3) There are also phenomena of *slow-down as an unintended consequence* of acceleration and dynamization. This frequently entails *dysfunctional* and *pathological* forms of deceleration; the best known version of the former is the traffic jam, whereas recent scientific findings identify the latter in some forms of psycho-pathological depression that are understood as individual (deceleratory) reactions to overstretched pressures of acceleration.³⁷ This category could also include the structural exclusion of workers from the sphere of production, which is often a consequence of their inability to keep up with the flexibility and speed required in modern Western economies. The excluded thus suffer extreme ‘deceleration’ in the form of long-term unemployment.³⁸ Economic recessions – called economic slow-downs – could also be interpreted along these lines.
- 4) Contrary to the unintentional forms of slow-down there are *intentional forms of (social) deceleration* which include ideological movements against modern acceleration and its effects. Such movements have accompanied more or less every new step in the history of modern acceleration, and in particular of *technological acceleration*. Thus, the steam engine, the railway, the telephone, and the computer were met with suspicion and even hostility; in all cases, the oppositional movements eventually failed.³⁹ Hence, within this fourth category, we need to distinguish between two forms of deliberate deceleration:
 - a) On the one hand, there are limited or temporary forms of deceleration which aim at preserving the capacity to function and further accelerate

within acceleratory systems. On the individual level, we find such *accelerating forms of deceleration* where people take 'time out' in monasteries or take part in yoga courses which promise 'a rest from the race' – for the purpose of allowing a more successful participation in acceleratory social systems afterwards. Similarly, there is a huge self-help literature suggesting a deliberate slow-down in work or learning in order to increase the volume of overall work or learning in a given period of time, or recommending pauses in order to increase energy and creativity.⁴⁰ On the social and political level, too, 'moratoria' are sometimes suggested to solve technological, political, legal, environmental, or social obstacles that stand in the way of modernization.⁴¹

- b) On the other hand, there are diverse, often fundamentalist, anti-modernist social movements for (radical) deceleration. This is hardly surprising given the fact that acceleration appears to be one of the fundamental principles of modernity. Among these we find radical religious as well as 'deep ecological' or politically ultra-conservative or anarchist movements. Thus, for German politician and scholar Peter Glotz deceleration has become the new ideological focus of the victims of modernization.⁴²

However, to straightforwardly dismiss the cry for deceleration as ideological is dangerously simplistic, for the more important arguments for intentional deceleration are those that follow the lines of thought of the first form (4a). The central insight here is that the enormous processes of acceleration which have shaped modern society were firmly grounded and enabled by the stability of *some* central modern institutions like law, democracy, the industrial work regime, and the standardized or 'institutionalized' biographies or 'life-trajectories' of modernity.⁴³ Only within a stable framework formed by such institutions can we find the necessary preconditions for long-term planning and investment and thus for long-term acceleration. Furthermore, as Lübke argues, the preconditions of cultural reproduction in an accelerating society are such that flexibility is only possible on the basis of some stable and unchanging cultural orientations and institutions. Institutionally as well as individually – or structurally as well as culturally – there seem to be certain limits to flexibilization and dynamization which may be in danger of erosion in late modernity.⁴⁴ Hence, it is quite possible that, much more than the anti-modernist radicals, it is the very success and ubiquity of acceleration that undercuts and erodes the preconditions for future acceleration. In this sense, deceleration in some respects could be a functional necessity of acceleration society rather than an ideological reaction to it.

- 5) Finally, we find the perception that in late-modern society, despite widespread acceleration and flexibilization which create the appearance of total contingency, hyper-optionality, and unlimited openness, 'real' change is in fact no longer possible: the system of modern society is closing in and history is

coming to an end in a 'hyper-accelerated standstill' or 'polar inertia.' Advocates of this diagnosis include Paul Virilio, Jean Baudrillard, and Francis Fukuyama. They claim that there are no new visions and energies available to modern society and hence the enormous speed of events and alterations is a superficial phenomenon barely covering up deep-rooted cultural and structural inertia.⁴⁵ For a sociological theory of acceleration society, it is vital to account for this possibility of (extreme) paralyzation in its very conceptual scheme.

The fundamental question that arises at this point is the relationship between processes of social acceleration and deceleration in modern society. Two general possibilities are conceivable. First, the processes of acceleration and deceleration are by and large in balance such that we find both types of changes in the temporal patterns of society without a clear and sustained dominance of one or the other. Second, the balance in fact shifts towards the powers of acceleration such that the categories of deceleration have to be interpreted either as *residual* or as *reactions* to acceleration. I would suggest that the second is in fact correct, though this is rather difficult to prove empirically. My claim rests on the supposition that none of these forms of deceleration amounts to a genuine and structurally equal counter-trend to modern acceleration. The phenomena listed under categories (1) and (2) merely denote the (retreating) limits of social acceleration; they are not counter-powers at all. The decelerations of category (3) are *effects* of acceleration and as such derivative of, and secondary to, it. Category (4a) identifies phenomena which, on closer examination, turn out to be either elements of acceleration processes or *enabling conditions* of (further) acceleration. The intentional resistance to the speeding up of life and the ideology of deceleration (4b) is clearly a *reaction* to pressures of and for acceleration; as was pointed out above, all of the main tendencies of modernity have met considerable resistance, but so far all forms of resistance have turned out to be rather short-lived and unsuccessful. Thus, the only form of deceleration that seems not to be derivative or residual is category (5). This dimension seems to be an inherent, complementary feature of modern acceleration itself; it is the paradoxical flipside characteristic of all the defining forces of modernity (individualization, differentiation, rationalization, domestication, and acceleration).

V. Ethical and Political Implications

It is of central importance for the analysis of the temporal structures of society to realize that the speeding-up of processes frequently amounts to more than a mere quantitative change that leaves the nature of those processes untouched. To the contrary, just as speeding up a sequence of images can 'bring them to life' in the transition from photography to film, or the acceleration of molecules can transform ice into water into steam, changes in the temporal structures of modern societies transform the very essence of our culture, social structure, and personal

identity (and, of course, our experience of nature, too). Thus, the much discussed but little agreed upon distinction between modernity and post- or late modernity may in fact be best captured with reference to the temporal dimension. *Late modernity is nothing other than modern society accelerated (and desynchronized) beyond the point of possible reintegration.* I would like to show this by exploring two major and related transformations: the transition in personal identities and the decline of politics in late modernity.

1) Situational Identity and the De-Temporalization of Life

Of the three dimensions of social acceleration, the *acceleration of the pace of life* is most directly linked to personality. Since the notion of 'personality' has become rather obscure in the social sciences and humanities (probably owing to its essentialist ring), the relevant changes are instead most often discussed in terms of shifting patterns of *identity*.⁴⁶

From the model developed above, the acceleration of the *pace of life* could be explained by two different factors. On the one hand, individuals could feel pressed to speed up in response to the social change around them, i.e., because of what I have called the 'slippery slope' phenomenon. Acceleration in this sense would be enforced by the fear of losing out in light of the speed and flexibility demands of the social and economic world. On the other hand, speeding up the pace of life could be a (voluntary) response to 'the promise of acceleration,' i.e., a consequence of people's conception of the good life. Of course, *fear* and *promise* might well *both* be driving factors of acceleration (as they are, following Max Weber's famous Protestant ethic thesis, the driving factors of capitalism).⁴⁷ Common parlance might serve as a guide for testing the slippery-slope hypothesis: even a casual look at how people explain or justify their use of time is puzzling in light of the dominant ideology of individual freedom. In a strange opposition to the idea that individuals in Western societies are free to do whatever they please, the rhetoric of obligation abounds: "I really have to read the newspaper, exercise, call and visit my friends regularly, learn a second language, scrutinize the market for job opportunities, have hobbies, travel abroad, keep up with contemporary computer technologies, etc."⁴⁸ However, we have to be careful to scrutinize the source of this kind of 'must' – it might well stem from an underlying cultural ideal as well as from social and economic pressures. Interestingly, we find an analogous semantic slip in contemporary politics: whereas in early and 'classical' modernity processes and technologies of acceleration were legitimized in the rhetoric of 'progress,' which mirrored the *promise of acceleration*, in late modernity political language has adopted the terminology of 'inherent necessity' and 'unavoidable adjustment' (to a fiercely competitive world) – a clear indication of the felt pressures of the 'slippery slope.'

However, common parlance might serve as an indicator for the nature of subtle connections between the different spheres of social acceleration in yet another

sense. The acceleration of rates of social change to an *intra-* rather than *inter-*generational pace is mirrored in a language which avoids identity predicates and uses temporary markers instead. People speak of *working* (for the time being) as *a baker* rather than *being a baker*, *living with Mary* rather than *being Mary's husband*, *going to the Methodist Church* rather than *being a Methodist*, *voting Republican* rather than *being a Republican*, and so on. This use of language indicates that the awareness of *contingency* has increased even where the actual rates of change have not yet done so: things (jobs, spouses, religious and political commitments, etc.) *could* be otherwise, they *could change* at any time because of either my own or other people's decisions. Although increased contingency is not equivalent to acceleration, it surely contributes to the perception of slippery slopes and time-pressure. The introduction of temporary markers in identity statements (I *was then* a Methodist, I *am now* married to Mary, I *will be* a consultant *after* my next degree) thus reflects a temporal 'contraction of identity' reflecting the 'contraction of the present' identified above. It is measurable, to some degree, by indicators for the de-institutionalization of biographies and life-trajectories.

Accordingly, a number of recent studies suggest a significant change in the time perspectives by which people organize their lives. As Martin Kohli has convincingly argued, modernity was characterized by a "temporalization of life": people were no longer absorbed with handling their life on a day-to-day basis but started to conceive their lives along the lines of a three-tiered temporal pattern (the modern standard biography of *education*, *work-life*, *retirement* or *childhood*, *adult life*, *old age*) that defined an institutionalized, reliable structure and orienting perspective around which individuals could plan their lives.⁴⁹ 'Classical' modern identities were consequently long-term projects supposed to evolve like a *Bildungsroman*. In late modernity, however, this pattern no longer holds: neither work- nor family-life can be foreseen or planned for a lifetime. Instead, people develop a new perspective that has been oddly termed the "temporalization of time": time-spans and the sequence and duration of activities or commitments are no longer planned ahead but left to evolve.⁵⁰ Such a 'temporalization of time,' however, is equivalent to the *de-temporalization of life*: life is no longer planned along a line that stretches from the past into the future; instead, decisions are taken from 'time to time' according to situational and contextual needs and desires. As Richard Sennett argues in his well-known essay on *The Corrosion of Character*, stability of character and adherence to a time-resistant life-plan are incompatible with the demands of the late-modern world.⁵¹ Thus, a conception of the good life based on long-term commitments, duration, and stability is thwarted by the fast pace of social change.⁵²

But even when this new perspective is depicted in neutral or even positive terms, it is evident that a new form of 'situationalism' is replacing the temporally extended identity characteristic of classical modernity.⁵³ This 'new situationalism' in a way resembles premodern forms of existence in which people had to cope with unforeseeable contingencies on a day-to-day basis without being able

to plan for the future; however, whereas the dangers, events, and contingencies that threatened *their* forms of life (natural disasters, wars, diseases, etc.) were exogenous to society, the 'new situationalism' is an endogenous product of social structures themselves. However we evaluate this phenomenon, the incompatibility of 'situational' identities with the modern ideal of individual ethical autonomy is apparent. For the ideal of the autonomous and reflective *leading* of a life requires adopting long-term commitments which bestow a sense of direction, priority, and 'narratability' to life.⁵⁴

The incapacity to engage in long-term commitments and to develop a frame of time-resistant priorities and long-term goals frequently seems to lead to a paradoxical backlash in which the experience of frantic change and 'temporalized time' gives way to the perception of 'frozen time' without (a meaningful) past and future and consequently of depressing inertia. German philosopher Klaus-Michael Kodalle has tried to explain this phenomenon philosophically, while Douglas Coupland's *Generation X* illustrates it metaphorically in the stories of 'Texlahoma,' a place in which time is eternally frozen in the year 1974 – making for a nice contrast with the book's subtitle: 'Tales for an Accelerated Culture.' Finally, Peter Conrad observes that historically, the problem of *l'ennui* (boredom) becomes vexing precisely at the moment when the Industrial Revolution "increased velocity in all areas of human experience" and created a climate of "hectic, propulsive dynamism" in which history itself was imagined as a fast-running railway.⁵⁵

In sum, the individual's reaction to social acceleration in late modernity seems to result in a new, situational form of identity, in which the dynamism of 'classical' modernity, characterized by a strong sense of direction (perceived as *progress*), is replaced by a sense of directionless, frantic motion that is in fact a form of inertia.

2) The 'End of Politics' and the De-Temporalization of History

Interestingly, an exactly analogous phenomenon can be observed in late-modern politics. Here we have the same constellation of a *political* 'temporalization of time' which results in a de-temporalization of politics. Politics in 'classical modernity' had a temporal index in the very labels 'progressive' vs. 'conservative' (or left vs. right), while history was perceived of as (directed) progress: *progressive* politics sought to accelerate this historical movement, whereas *conservative* politics was 'reactionary' in opposing the forces of change and acceleration. Today, ironically, if the distinction between left and right has retained any discriminatory power at all, 'progressives' tend to sympathize with the advocates of deceleration (stressing locality, political control of the economy, democratic negotiation, environmental protection, etc.), whereas 'conservatives' have become strong defenders of the need for further acceleration (embracing new technologies, rapid markets, and fast administrative decision-making). This

is yet another example of how the forces of acceleration have outgrown the very agents and institutions which set them in motion: bureaucracy, the nation-state, the strict time-regime of the factory, democratic politics, stable personal identities – all of these institutions historically played a key role in enabling social acceleration by providing stable and calculable background conditions, but are now in danger of being eroded by the very forces of acceleration they set in motion.⁵⁶ In late modernity, they have become obstacles to further acceleration. The very idea of an ‘institution,’ whose Latin root indicates its static, durable character, is incompatible with the idea of ‘total’ acceleration (Figure 3).

As a result, politics, too, has become ‘situationalist’: it confines itself to *reacting* to pressures instead of developing progressive visions of its own. Very often, political decisions no longer aspire to actively steer (acceleratory) social developments, but are defensive and deceleratory. It seems that just as it has become virtually impossible to individually plan one’s life in the sense of a ‘life-project,’

Figure 3: The Dialectics of Acceleration and Institutional Stability:
Modern Accelerators as Late-modern Decelerators

Societal Core Institutions	As Accelerators in ‘Classical Modernity’	As Decelerators in ‘Late-Modernity’
Bureaucracy	Acceleration of administration processes	Deceleration of social and economic processes
Nation-State	Acceleration via standardization (time, language, law)	Deceleration of supra-national circulatory processes
Representative Democracy	Accelerated adaptation to sociopolitical needs	Deceleration of vital decision-making
Political Regulation	Acceleration through ‘progressive’ politics	Deceleration through the claim to regulation
Spatial and temporal separation of work and ‘life’/leisure	Uninhibited acceleration of productive economic processes	Inhibited acceleration of the lifeworld
Stable Personal Identities	Acceleration through individualization	Deceleration/inhibited change through inflexibility
Individual Life-Plans	Acceleration through the temporalization of life	Decelerated adaptation to social change

it has become politically impossible to plan and shape society over time; the time of political projects, it seems, is also over. Individually as well as politically, the sense of a directed movement of history has given way to a sense of directionless, frantic change. For Armin Nassehi, a German author in the systems theoretic tradition, this loss of political autonomy (corresponding to the loss of individual autonomy discussed above) is an inevitable consequence of the temporal structures of modern society:

The present . . . loses its capacity for planning and shaping. As the present of action it is always oriented towards the future, but it cannot shape this future because of the dynamics, risks, and vast amount of simultaneity within the present, which it cannot control at all. Early modernity promised the capacity to shape and control world and time and to initiate and historically legitimate future progress. But in late modernity, time itself has come to destroy the potential for any form of social or substantial control, influence, or steering.⁵⁷

The structural problem at the heart of this *disappearance of politics* is the political system's fundamental inability to accelerate. Here we touch on a central structural feature of late-modern societies: the desynchronization of social and functional spheres, which takes two forms. First, there is a desynchronization of different groups and segments of society. Not all social groups accelerate equally: some, like the sick, the unemployed, the poor, or, in some respects, the elderly, are forced to 'decelerate,' while others, like the Amish, refuse to adopt the temporal structures and horizons of modernity. This desynchronization entails an increasing 'simultaneity of the non-simultaneous': high-tech and stone-age methods of warfare, transport, or communication persist side by side, not only between different countries, but even within the same society, and fast and slow paces of life can be observed on one and the same street.⁵⁸ The result of this 'multitemporality' is likely to be a progressive *disintegration* of society. At first, the desynchronization of various segments might aggravate the problem of 'ghettoization,' transforming society into a mosaic of temporal ghettos. Now some of these ghettos might resist the forces of acceleration, but wherever these forces are in operation, they will eventually enforce the dissolution of the boundaries between groups and segments, since these boundaries are effective speed limits (the increasing irrelevance of state borders is just the most striking example of this tendency). The resulting postmodern de-differentiation, however, may not lead to reintegration, but to a fast-paced, atomized, kaleidoscopic social amalgam in which highly volatile associations and lifestyle milieus replace the 'mosaic of ghettos.'

This, in turn, may aggravate the political problem of desynchronization in its second form. Contrary to a widespread view, modernity has not just established a single, unitary form of abstract, linear time that synchronizes its various subsystems. Rather, the process of functional differentiation resulted in a series of almost autopoietic subsystems like the economy, science, law, politics, the arts,

etc., all of which follow their own temporal rhythms, patterns, and horizons. Just as there is no unifying social or substantial center governing the subsystemic operations, there is also no integrating temporal authority, and this, in turn, results in increasing temporal desynchronization.

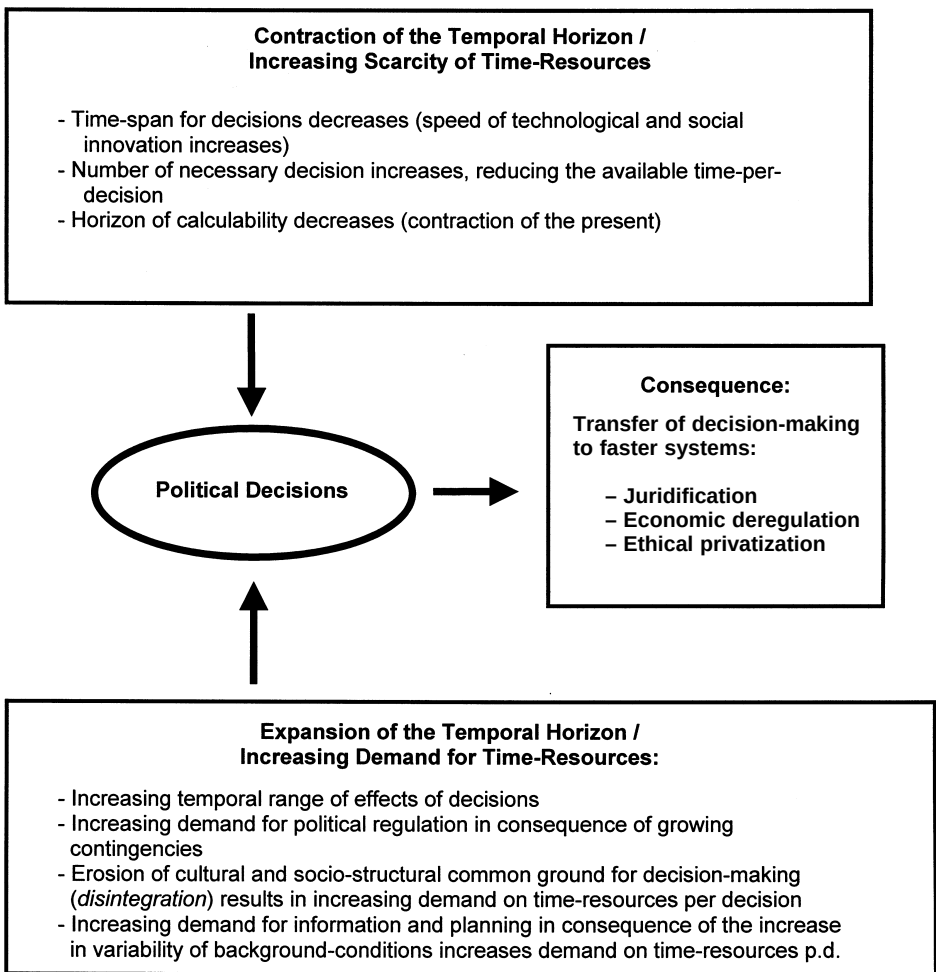
For the political system, this entails truly paradoxical temporal horizons. On the one hand, the time needed for democratic political decision-making is not just hard to accelerate, since processes of deliberation and aggregation in a pluralistic democratic society inevitably take time;⁵⁹ it is actually *increasing* for a number of reasons. First, the less consensus there is within society, the less conventionalist the legitimating principles of society are, the longer it takes to reach consensus – and ‘disintegrated’ modern acceleration societies tend to become both more pluralistic and less conventionalist, making it hard to know in advance even which social groups or associations are going to be relevant for negotiations. Hence, in a volatile political world, the time needed for the effective organization of collective interests increases.⁶⁰ Second, the less certainty there is about future conditions, the longer it takes to plan for the future and to make decisions.⁶¹ Due to the acceleration of social change and the ‘contraction of the present,’ background conditions become increasingly contingent; instead of providing yardsticks for decision-making, they become complicating factors. Third, the *effects* of political decisions tend to extend further and further into the future – most visibly in the area of nuclear power or genetic engineering, where decisions seem to be irreversible. The longer the temporal range of a particular decision, the more time it takes to rationally make it. Here, perhaps, the paradoxical nature of policy making today is most visible: the effects of crucial decisions extend in time just as the time available for making them shrinks.

On the other hand, contrary to this need for *more time for political decision-making*, the acceleration of the surrounding systems – especially economic circulation and technological-scientific innovation – *decreases* the time given to politics to decide an issue. If politics aspires to steer and control the basic conditions of technological and economic development, it has to either keep up with their accelerating pace or seriously infringe on their autonomy, virtually ending functional differentiation. At present, policy makers are always in danger of making completely anachronistic decisions: when, after years of deliberation and negotiation, they finally pass a law regulating the use of, say, some forms of stem-cell research or cloning, technological progress may already have made it obsolete.⁶² Second, because of the ‘contraction of the present’ and the increase in contingency, not just the rate, but also the number and range of social issues in need of political regulation rise, leaving less time for each decision.

Third, since, as we have seen, background conditions change quickly and the temporal horizon for which political effects can be rationally planned and controlled continuously contracts, fewer and fewer things can be durably and effectively regulated. Instead, politics shifts to ‘muddling through’ (described by Luhmann as the *primacy of the short-term*) with increasingly temporary and

provisional solutions, ensuring that issues keep reappearing on the agenda. The result of these contradictory and incompatible time pressures seems to be consistent with our finding above: politics not only becomes 'situationalist' and loses its sense of direction; it also tends to shift the decision-making process towards other, faster arenas: the legal system (juridification), or the economy and individual responsibility (privatization and deregulation). Thus, precisely at a point in history where the human power to steer and control its own fate seems to reach an unprecedented technological zenith (foremost, of course, in the shape of genetic engineering), society's political capacity to do so reaches its nadir. The deliberate and democratic political shaping of our society and form of life, the political project and promise of enlightened

Figure 4: Paradoxes of Political Time



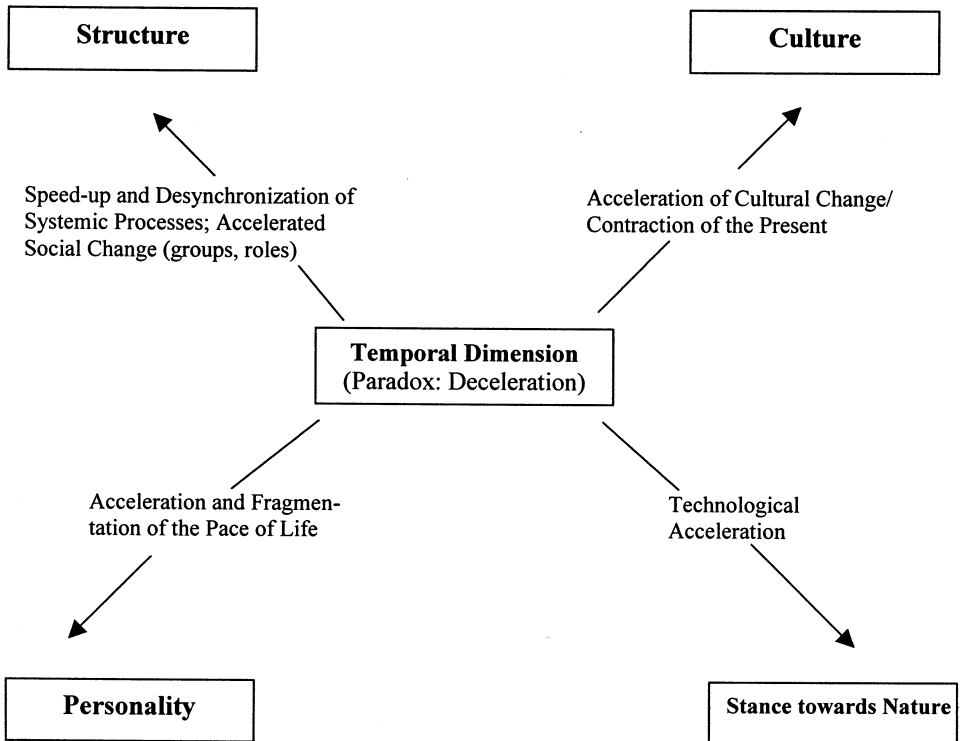
modernity, may thus be becoming obsolete in the late-modern 'acceleration society' (Figure 4).⁶³

As a result, the inability to control social change has brought an overwhelming sense of directionless change in an 'iron cage' that itself has become fundamentally inert. Parallel to the individual experience of time and life delineated above, the frantic pace of political 'events' covered in the news can barely cover up the virtual standstill of the history of ideas – or of history itself. As Baudrillard puts it:

This is the most significant event within [our] societies: the emergence, in the very course of their mobilization and revolutionary process (they are all revolutionary by the standards of past centuries), of an immense indifference and the silent potency of that indifference. This inert matter of the social is not produced by a lack of exchanges, information or communication, but by the multiplication and saturation of exchanges. . . . It is the cold star of the social and, around that mass, history is also cooling. Events follow one upon another, canceling each other out in a state of indifference. The masses, neutralized, mithridatized by information, in turn neutralize history and act as an *écran d'absorption*. . . . Political events already lack sufficient energy of their own to move us. . . . History comes to an end here, not for want of actors, nor for want of violence. . . , nor for want of events. . . , but by deceleration, indifference and stupefaction. . . . It is being buried beneath its own immediate effect, worn out in special effects, imploding into current events. Deep down, one cannot even speak of the end of history here, since history will not have time to catch up with its own end. Its effects are accelerating, but its meaning is slowing inexorably. It will eventually come to a stop and be extinguished like light and time in the vicinity of an infinitely dense mass. . . .⁶⁴

VI. Conclusion

At the outset of this paper, I tried to pinpoint the relevance of the logic of acceleration for the overall process of modernization. I noted that in the sociological tradition, modernization has been analyzed from four different perspectives relating to society's structure, culture, personality-type, and relation to nature. Now, when we try to reconceptualize the acceleration process along the lines of these four dimensions, it appears that change in the temporal structures is relevant to each of them (Figure 5). While all segments of the world's population are certainly not equally affected by social acceleration – to the contrary, in some parts of the world and some segments of Western societies many processes in fact appear to decelerate – the logic of social acceleration is decisive for the structural and cultural evolution of contemporary society. Thus, it is evident that *technological acceleration* is a crucial feature of modern society's relation to nature, while the *acceleration of the pace of life* is of overriding importance for the late modern personality. Furthermore, the overall *acceleration of social change* is intimately related to both cultural and structural transformation.

Figure 5: The Process of Modernization II

With respect to social structure, two distinct but related aspects of acceleration were identified. On the one hand, if we take functional differentiation to be the main structural feature of modern societies, there clearly is a speed-up of (sub-)systemic processing: financial transactions, economic production and distribution, scientific discoveries, technological inventions, artistic productions, and even law-making⁶⁵ have unmistakably sped up following their own logics without much outside interference. This has led to desynchronization, since not all subsystems are equally capable of acceleration.⁶⁶ On the other hand, if we take the basic structure of society to be the structure of its associations, groups, and collectivities and the corresponding role structures – as suggested, e.g., by Parsons⁶⁷ – then processes of social acceleration have clearly increased the speed of social change; family and occupational structures as well as associations and milieus have become highly volatile, shifting, and contingent, making it difficult to identify politically and socially relevant and stable associational structures at all. This, as we have seen, further aggravates the problem of social integration for late-modern societies.

With respect to the cultural dimension, the 'contraction of the present,' i.e., the

shortening of the time-spans within which action orientations and social practices remain stable, is the most important effect of social acceleration. Lifestyles, fashions, practices, occupational, familial, territorial, political, and religious commitments all change at a faster pace and become increasingly contingent and revisable. Late-modern culture thus clearly appears to be highly dynamic.

However, as was pointed out throughout this essay, in at least three of the four dimensions (structure, culture, and personality) we also find complementary signs of deceleration or inertia – the paradoxical flipside of social acceleration. Thus, individuals at times experience their loss of direction, priorities, and narratable ‘progress’ as ‘frozen time’ or virtual inertia despite the frantic pace of events, just as the entrenched logics of subsystemic operations appear to be so reified that notions of the ‘end of history,’ the ‘exhaustion of utopian energies,’ and the ‘iron cage’ abound amidst the discourse of permanent and total social change.⁶⁸ Similarly, with respect to culture, what from one perspective appears to be a vast contingency of value orientations and lifestyles in which ‘anything goes’ can be interpreted from another perspective as an entrenchment of the basic value orientations of modernity, i.e., as strict adherence to the values of activity, universality, rationality, and individuality.⁶⁹ From the ‘deceleratory perspective,’ Western societies’ apparently fast pace of change is only a surface phenomenon beneath which we find inertia. Only with respect to nature does there seem to be no complementary deceleration. Here, only the looming possibility of environmental disaster figures as a potential for (exogenous) deceleration.

A final question concerns whether acceleration really is an *independent feature* of modernity or just a *perspective* from which its core processes (individualization, domestication, rationalization, differentiation) can be reinterpreted. After all, all four processes traditionally associated with modernization are intricately connected with increases in speed: thus, individualization can be a cause as well as an effect of acceleration, since individuals are more mobile and adaptive to change and faster in making decisions than collectivities. Similarly, one of the main reasons for, as well as consequences of, organizational differentiation is the speeding up of systemic processes, and the same holds true for rationalization as the improvement of means-ends relationships and domestication as an improvement of instrumental control. Nevertheless, I would suggest that acceleration is an irreducible and constitutive trait of modernization for at least three reasons. First, individual as well as collective human existence is in its very essence temporal and processual; changes in temporal structures are changes in individual and social existence.⁷⁰ Hence, it is only with respect to the significant changes in its temporality that the nature and impact of modernization become fully visible. Second, social acceleration reveals the unitary logic underlying all four dimensions of modernization. And third, it is only from a temporal perspective that we can fully understand the fundamental transformations in contemporary society, which are the result of social acceleration *within* the unaltered framework of modernity but *beyond* the limits of individual and

social integration and autonomy. As such, they amount to a silent but sweeping qualitative social revolution by a mere quantitative change in the realm of speed.

NOTES

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1. James Gleick, *Faster: The Acceleration of Just About Everything* (New York: Pantheon, 1999).

2. Cf. William E. Scheuerman, "Liberal Democracy and the Empire of Speed," *Polity* (2001): 1: "Any attempt to make sense of the human condition at the start of the new century must begin with an analysis of time and space compression." 'Time and space compression' is David Harvey's term for acceleration; see *The Condition of Postmodernity: An Enquiry into the Origins of Cultural Change* (Cambridge, MA and Oxford: Blackwell, 1990).

3. Cf. Reinhart Koselleck, *Futures Past: On the Semantics of Historical Time* (Cambridge, MA: MIT Press, 1985), who demonstrates that complaints about the overwhelming speed of modern history start well before the French Revolution and long before there is a noticeable development of *technological* speed. The discourse on acceleration then periodically peaks again and again during the following centuries. For example, in 1877, W.G. Greg observed that the most significant feature of *his* age was its high speed and the pressure it put on life, and he voiced severe doubts whether this gain in speed was a good that was worth its price. In 1907, Henry Adams formulated his famous 'law of acceleration' (of history); cf. Robert Levine, *A Geography of Time* (New York: Basic, 1997) for further historical observations of social acceleration.

4. Thus, the protests and anxieties relating to the introduction of the steam engine, the railway, the telephone, or the PC mirror in many respects the various 'communitarian' anxieties and protests against manifestations of *individualization*, or traditionalistic opposition against ensuing waves of *rationalization* – with the latter generally being victorious in the process of modernization.

5. See Barbara Adam, *Time and Social Theory* (Oxford: Polity, 1990) for an attempt to systematically and comprehensively introduce the temporal dimension into social theory.

6. This conceptualization of sociological attempts to grapple with processes of modernization is suggested by Hans van der Loo and Willem van Reijen in *Modernisierung. Projekt und Paradox*, 2e (Munich: DTV, 1997), loosely adhering to Talcott Parsons' (notoriously static) outline of a *General System of Action*.

7. Paul Virilio, "Polar Inertia," in *The Virilio Reader*, ed. James Der Derian (Oxford: Blackwell, 1998), 117–33; Francis Fukuyama, *The End of History and the Last Man* (New York, 1992); Jean Baudrillard, *The Illusion of the End* (Oxford: Polity, 1994); Charles Taylor, "Marx statt Tocqueville. Über Identität, Entfremdung und die Konsequenzen des 11. September. An Interview with Hartmut Rosa and Arto Laitinen," *Deutsche Zeitschrift für Philosophie* 50, no. 1 (2002): 9.

8. Lothar Baier, *Keine Zeit! 18 Versuche über die Beschleunigung* (Munich 2000). This sense is strongly expressed in the novels of Douglas Coupland, too. I have explored this at length in "Am Ende der Geschichte: Die Generation 'X' zwischen Globalisierung und Desintegration," in *Neustart des Weltlaufs? Studien zum fin de siècle-Motiv in der Moderne*, ed. Karsten Fischer (Frankfurt/M.: Suhrkamp).

9. Manfred Garhammer, *Wie Europäer ihre Zeit nutzen. Zeitstrukturen und Zeitkulturen im Zeichen der Globalisierung* (Berlin: edition sigma, 1999), 378.

10. Karlheinz Geißler, *Vom Tempo der Welt. Am Ende der Uhrzeit* (Freiburg: Herder, 1999), 89.
11. Paul Virilio, *Open Sky* (London/New York: Verso, 1997), 9ff.
12. See e.g. Harvey, *The Postmodern Condition*, 201ff.
13. Marc Augé, *Non-Lieux. Introduction à une anthropologie de la surmodernité* (Paris 1992). Harvey, however, referring to an inverse spatialization of time (*The Postmodern Condition*, 272f), cautions us to not dismiss space too quickly.
14. See e.g., Georges Gurvitch, "Social Structure and the Multiplicity of Time," in E.A. Tiryakian, ed., *Sociological Theory, Values, and Sociocultural Change* (London: Free Press of Glencoe, 1963) or Gerhard Schmied, *Soziale Zeit. Umfang, 'Geschwindigkeit,' Evolution* (Berlin: Duncker & Humblot, 1985), 86ff.
15. Arjun Appadurai, "Disjuncture and Difference in the Global Cultural Economy," in *Global Culture: Nationalism, Globalization and Modernity*, ed. Mike Featherstone (London 1990), 295–310.
16. Cf. Piotr Sztompka, *The Sociology of Social Change* (Oxford: Blackwell, 1993) or Hans-Peter Müller and Michael Schmid, eds., *Sozialer Wandel. Modellbildung und theoretische Ansätze* (Frankfurt/M, 1995). In "Social Structural Time: An Attempt at Classifying Types of Social Change by Their Characteristic Paces" (*The Rhythms of Society*, ed. Michael Young and Tom Schuller (London and New York: Routledge, 1988), pp.17–36), Peter Laslett distinguishes between nineteen (!) different rates of internal social change (economic, political, cultural, etc.).
17. Hermann Lübke, "Gegenwartsschrumpfung," in *Die Beschleunigungsfalle oder der Triumph der Schildkröte*, ed. Klaus Backhaus and Holger Bonus, 3e (Stuttgart: Schäffer/Pöschel, 1998), 129–64.
18. Cf. Laslett "Social Structural Time," 33.
19. For some empirical evidence on this, cf. Garhammer, *Wie Europäer ihre Zeit nutzen* and Richard Sennett, *The Corrosion of Character: The Personal Consequences of Work in the New Capitalism* (New York: Norton, 1998).
20. Peter Wagner, *A Sociology of Modernity: Liberty and Discipline* (London: Routledge, 1994); Ulrich Beck, Anthony Giddens, and Scott Lash, *Reflexive Modernization: Politics, Tradition and Aesthetics in the Modern Social Order* (Cambridge: Polity, 1994).
21. Georg Simmel, "The Metropolis and Mental Life," *On Individuality and Social Form*, ed. D. Levine (Chicago 1971) and *The Philosophy of Money*, ed. David Frisby (London: Routledge, 1978), 470–512; Levine, *A Geography of Time*.
22. For a very interesting economic explanation, see Staffan B. Linder, *The Harried Leisure Class* (New York: Columbia University Press, 1970). I take up some of his arguments in the following.
23. American sociologist Robert Levine and his team recently conducted a cross-cultural comparative empirical study in which three indicators for the speed of life were used: the *speed of walking* in inner-cities; the *time it takes to buy a stamp* in a post-office; and the *exactness of public clocks*. For a number of reasons I have discussed at length elsewhere, this approach can at best serve as a very rough preliminary attempt. It certainly remains very unsatisfactory as an instrument in a thorough sociological analysis of the temporal structures of late modernity. Rosa, "Temporalstrukturen in der Spätmoderne: Vom Wunsch nach Beschleunigung und der Sehnsucht nach Langsamkeit. Ein Literaturüberblick in gesellschaftstheoretischer Absicht," *Handlung, Kultur, Interpretation* 10 (2001).
24. Geißler, *Vom Tempo der Welt*, 92; Garhammer, *Wie Europäer ihre Zeit nutzen*, 448ff; Levine, *A Geography of Time*, 196f.
25. It thus remains rather doubtful that John P. Robinson and Geoffrey Godbey's diagnosis of the beginning of "The Great American Slowdown" (*American Demographics* (June 1996): 42–48) can be confirmed by further investigation. As of January 2002, Robinson confirmed to the author by personal communication that he doesn't have conclusive evidence in either direction yet.
26. Cf. Garhammer, *Wie Europäer ihre Zeit nutzen*.
27. See also Friederike Benthous-Apel, *Zwischen Zeitbindung und Zeitautonomie. Eine empirische Analyse der Zeitverwendung und Zeitstruktur der Werktags- und Wochenendfreizeit* (Wiesbaden: Deutscher Universitäts Verlag, 1995).

28. Cf. Sherry Turkle, *Life on the Screen: Identity in the Age of the Internet* (New York: Simon & Schuster, 1995).

29. Thus, elderly people in Western society are frequently unable to understand the 'techno-bubble' the young use when talking about their gameboys, emails, DVDs, etc.

30. Cf. Sennett, *The Corrosion of Character*; Fritz Reheis, *Die Kreativität der Langsamkeit. Neuer Wohlstand durch Entschleunigung*, 2e (Darmstadt: Primus, 1998); Harvey, *The Postmodern Condition*; cf. Scheuerman "Liberal Democracy and the Empire of Speed," 2ff. See Moishe Postone, *Time, Labor, and Social Domination: A Reinterpretation of Marx's Critical Theory* (Cambridge: Cambridge University Press, 1996) for a contemporary Marxist reinterpretation of the roles of growth and time in capitalist societies.

31. The clearest expression of the 'eudaimonism of speed' probably is Marinetti's Manifesto of Futurism, where *speed* is celebrated as a glorious new, omnipresent, and immortal goddess.

32. "Everything has begun to move, or has been set in motion, and with the intention or under the pretense of fulfilling and completing everything, everything is placed in question, doubted, and approaches a general transformation. The love of movement in itself, without purpose and without specific end, has emerged and developed out of the movement of the time. In it, and in it alone, one seeks and sets real life" (quoted in Kosselleck, *Futures Past*, 251). It is remarkable how closely these observations resemble Marx and Engels' famous passage from the *Communist Manifesto* that *all that is solid melts into air*. However, whereas in the *Manifesto* this appears to simply be an (unintended) byproduct of economic relations, Ancillon (and Kosselleck) point out that this transformation is also driven by cultural currents. The question of the 'prime mover' can safely be set aside at this point.

33. Hans Blumenberg, *Lebenszeit und Weltzeit* (Frankfurt/M.: Suhrkamp, 1986); Marianne Gronemeyer, *Das Leben als letzte Gelegenheit. Sicherheitsbedürfnisse und Zeitknappheit*, 2e (Darmstadt: Wissenschaftliche Buchgesellschaft, 1996); Schulze, "Das Projekt des schönen Lebens. Zur soziologischen Diagnose der modernen Gesellschaften," in *Lebensqualität. Ein Konzept für Praxis und Forschung*, ed A. Bellebaum and K. Barheier (Opladen, 1994). On the secularization of time, cf. Charles Taylor, "Modern Social Imaginaries" (manuscript, 2001), 30ff.

34. Famous literary illustrations of this idea can be found in the work of Goethe, e.g., in *Faust* or *Wilhelm Meister*. Unsurprisingly, as Manfred Osten points out, Goethe's writings can fruitfully be read and interpreted as a description and critique of acceleration society; see "Accelerated Time: A Few Remarks on the Modernity of Goethe" (manuscript, 2002).

35. Of course, this multiplication of options is only a cultural problem if the options are considered to be (at least potentially) valuable. An increase in uninteresting options would not affect the pace of life. However, as we will see in section four of this paper, one problem of 'acceleration societies' is that the (future) relevance of present options becomes increasingly unpredictable. *We don't know what we will need, like, own and use tomorrow*.

36. Otthein Rammstedt, "Alltagsbewußtsein von Zeit," *Kölner Zeitschrift für Soziologie und Sozialpsychologie* 27 (1975): 50; cf. Niklas Luhmann, *The Differentiation of Society*, tr. Stephen Holmes and Charles Larmore (New York: Columbia University Press, 1982); Armin Nassehi, *Die Zeit der Gesellschaft. Auf dem Weg zu einer soziologischen Theorie der Zeit* (Opladen: Westdeutscher Verlag, 1993).

37. Cf. Levine, *A Geography of Time*, several articles in *Psychologie Heute* 26, no. 3 (1999), or Baier, *Keine Zeit!*, 147ff.

38. Sennett, *The Corrosion of Character*, 159ff; Marie Jahoda, "Time: A Social Psychological Perspective," in *The Rhythms of Society*, ed. Michael Young and Tom Schuller (London and New York: Routledge, 1988), 154–72.

39. Cf. Levine, *A Geography of Time*; Wolfgang Schivelbusch, *Geschichte der Eisenbahnreise. Zur Industrialisierung von Raum und Zeit im 19. Jahrhundert*, new ed, (Frankfurt/M.: Fischer Taschenbuch Verlag, 2000).

40. For such accelerating forms of deceleration, see Lothar J. Seiwert, *Wenn Du es eilig hast, gehe langsam: Das neue Zeitmanagement in einer beschleunigten Welt*, 5e (Frankfurt/New York: Campus, 2000).

41. Cf. Matthias Eberling, *Beschleunigung und Politik* (Frankfurt/M. 1996).

42. Peter Glotz, "Kritik der Entschleunigung," in Backhaus and Bonus, eds., *Die Beschleunigungsfälle oder der Triumph der Schildkröte*, 75–89, Cf. Levine, *A Geography of Time*.

43. See Rosa, "Temporalstrukturen in der Spätmoderne"; Martin Kohli, "Lebenslauf und Lebensalter als gesellschaftliche Konstruktionen: Elemente zu einem interkulturellen Vergleich," in *Im Lauf der Zeit. Ethnographische Studien zur gesellschaftlichen Konstruktion von Lebensaltern*, ed. G. Elwert, M. Kohli, and H.K. Müller (Saarbrücken 1990), 11–32; Holger Bonus, "Die Langsamkeit der Spielregeln," in *Die Beschleunigungsfälle oder der Triumph der Schildkröte*, ed. Backhaus and Bonus, 41–56.

44. Cf. Lübke, "Gegenwartsschrumpfung"; see also above, section II.2. and below, section IV.2.

45. Fukuyama, in fact, takes part in the discourse on 'post-histoire' that follows a long tradition going back to Kojève and Hegel. For Lothar Baier in his 'eighteen essays on acceleration' (*Keine Zeit!*), acceleration and change only happen on the 'user interface' of modern societies while their deep structures remain unchanged.

46. Cf. my *Identität und kulturelle Praxis. Politische Philosophie nach Charles Taylor*, pref. Axel Honneth (Frankfurt/M. and New York: Campus, 1998).

47. Cf. "Wachstum und Beschleunigung – Angst und Verheißung der kapitalistischen Gesellschaft," commentary on Elmar Altvater, *Kapitalismus. Zur Bestimmung, Abgrenzung und Dynamik einer geschichtlichen Formation*, in *Erwägen Wissen Ethik* (former *Ethik und Sozialwissenschaften*) 13, no. 3 (2002): 330–33.

48. For an impressive list of such 'musts' see Horst W. Opaschowski, *Freizeitökonomie: Marketing von Erlebniswelten*, 2e (Opladen: Westdeutscher Verlag, 1995), 85f.

49. Kohli, "Lebenslauf und Lebensalter"; cf. section II.2. above.

50. Cf. Mike Sandbothe, *Die Verzeitlichung der Zeit. Grundtendenzen der modernen Zeitdebatte in Philosophie und Wissenschaft* (Darmstadt: Wissenschaftliche Buchgesellschaft, 1998) and Karl H. Hörning, Daniela Ahrens, and Anette Gerhard, *Zeitpraktiken. Experimentierfelder der Spätmoderne* (Frankfurt/M.: Suhrkamp, 1997).

51. See, e.g., Hörning, Ahrens, and Gerhard's (ibid.) characterization of the late modern figure they term 'the gambler'; this character resembles the 'drifter' described by Richard Sennett, although Sennett is much more critical with respect to the desirability of such a time perspective. For a philosophical critique of such a 'reduced' sense of time that exclusively focuses on the present, see Dieter Sturma, "Die erweiterte Gegenwart. Kontingenz, Zeit und praktische Selbstverhältnisse im Leben von Personen," in *Die Wiederentdeckung der Zeit. Reflexionen, Analysen, Konzepte*, ed. A. Gimmler, M. Sandbothe, and W.Ch. Zimmerli (Darmstadt: Wissenschaftliche Buchgesellschaft, 1997), 63–78. For a further discussion of the causes and consequences of the 'temporalization of time,' see Rosa, "Temporalstrukturen in der Spätmoderne."

52. For this, see my "On Defining the Good Life: Liberal Freedom and Capitalist Necessity," *Constellations* 5 (1998): 201–14.

53. Cf. Hörning, Ahrens, and Gerhard, *Zeitpraktiken*.

54. Interestingly, the findings of some authors suggest that there is a tendency towards a 'new fatalism' which assumes that we cannot control or plan the conditions of our lives (e.g. Garhammer, *Wie Europäer ihre Zeit nutzen*). This, too, could be read as evidence for the increasingly 'situational' instead of 'temporal' perspective. Hörning, Ahrens, and Gerhard also point out that the incapacity to develop a persistent sense of relevance, direction, and priority is a serious problem for the late modern 'gambler' personality type (*Zeitpraktiken* and "Do Technologies Have Time? New practices of time and the transformation communicative technologies," *Time & Society* 8 (1999): 293–308).

55. Peter Conrad, *Modern Times, Modern Places* (New York: Alfred A. Knopf, 1999), 16f; Klaus-Michael Kodalle, ed., *Zeit-Verschwendung. Ein Symposium* (Würzburg 1999), 12; cf. Garhammer, *Wie Europäer ihre Zeit nutzen*, 482, Sennett, *The Corrosion of Character*, 121; at length Rosa, "Am Ende der Geschichte."

56. Thus, the neoliberal cry for deregulation is clearly a reaction to the perceived slowness of a bureaucracy that was once hailed by Max Weber as the fastest and most efficient institution

conceivable. Similarly, the power of the nation-state seems to be on the decline because its reactions are too slow in a world of high-speed global transactions. The slowness of the state was arguably a key factor in the fall of the Soviet Empire.

57. Nassehi, *Die Zeit der Gesellschaft*, 375. Similarly, Baudrillard holds that history has lost its sense of (linear) progression: it is no longer moving toward an imaginary end-point (which is why Baudrillard argued the 'symbolic year 2000' would never be reached), but rather "is disintegrating into its simple elements in a catastrophic process of recurrence and turbulence" (*The Illusion of the End*, 11). Cf. Baier, *Keine Zeit!*, 17f.

58. Of course, the acceleration thesis presented in this paper can only be upheld if it can be shown that the predominance of processes of deceleration and time abundance (if they can be found at all) is a phenomenon characteristic of socially excluded or underprivileged groups like the old, the unemployed and the poor or sick rather than the trend-setting elites. Deceleration in this sense is a sign of deprivation and exclusion and thus plays a dysfunctional rather than a functional role in modern societies, which can partially be explained as an unintended effect in the sense of category three above.

59. Cf. at length Scheuerman, "Liberal Democracy and the Empire of Speed," 9ff. Scheuerman points out that there is a long-standing trend in politics towards shifting political powers from the legislative to the executive branches of government in order to accelerate decision-making processes and to retain political control in the 'empire of speed.'

60. "How can a public be organized, we may ask, when literally it does not stay in place? . . . Without abiding attachments associations are too shifting and shaken to permit a public readily to locate and identify itself," John Dewey noted in 1927 in *The Public and Its Problems* (Athens: Ohio University Press, 1954), 140f; cf. Scheuerman, "Liberal Democracy and the Empire of Speed," 23ff.

61. Thus, it might be difficult to develop a good pension plan even in the face of stable demographics when the proportions of old age pensioners to working people is known. But this task is almost impossible when it is uncertain (a) how long future generations will live, (b) how long they will work, (c) what percentage of the population will work at what time, (d) whether the insurance system will be privatized anyway, (e) whether the nation-state will lose its power to decide over pension schemes altogether, etc.

62. This was the argument of the supporters of a Swiss proposition called the 'acceleration initiative' (*Beschleunigungsinitiative*), which sought to reduce democratic processes in order to speed up law-making. It was rejected in a March 2000 referendum.

63. Cf. Scheuerman, "Liberal Democracy and the Empire of Speed," 32f. Scheuerman, however, is still somewhat optimistic about the possibility of resynchronizing socioeconomic and political processes by way of institutional reform; cf. Scheuerman "Reflexive Law and the Challenges of Globalization," *Journal of Political Philosophy* 9, no. 1 (2001): 81–102.

64. Baudrillard, *The Illusion of the End*, 3f.

65. Cf. Carl Schmitt, *Die Europäische Rechtswissenschaft* (Tübingen: Universitätsverlag, 1950), 18–21 and 29–32; see also Scheuerman "The Economic State of Emergency," *Cardozo Law Review* 21 (2000): 1891ff.

66. It remains an open question whether at some point in the future the drive for further acceleration will cause an eventual dedifferentiation, as some advocates of 'postmodernity' argue, since strict systemic boundaries might eventually become obstacles to 'total' acceleration. If this should happen, functional differentiation could be a further feature in the dialectics of acceleration and stability (Figure 3): it *accelerated* social processes in 'classical' modernity but comes to function as a *decelerator* in late modernity. Some authors suggest that postmodernity has to be understood as a process of dedifferentiation in which the borders between science and religion, art and technology, economy and politics, etc. are crumbling; see Harvey, *The Condition of Postmodernity*, 291; Fredric Jameson, *The Cultural Turn: Selected Writings on the Postmodern 1983–1998* (London/New York: Verso, 1998).

67. Talcott Parsons, *The System of Modern Societies* (Englewood Cliffs: Prentice Hall, 1971), 4ff.

68. Even if there is eventually a postmodern dedifferentiation of social subsystems, this will, in my view, not alter the perception of inertia and powerlessness, since the resulting 'kaleidoscopic chaos' will be even less controllable. The operative blundering of systemic borders will not reverse the uncontrollable, non-intentional functioning of systemic processing.

69. For this argument, see Gerd Günter Voß, "Wertewandel – eine Modernisierung der protestantischen Ethik," *Zeitschrift für Personalforschung* 3 (1990): 263–75 and Schulze, "Steigerungsglogik und Erlebnisgesellschaft," *Politische Bildung* 30, no. 2 (1997): 79.

70. "The self is process; the temporal dimension is fundamental. To neglect the temporal dimension is to neglect the essence. We shall never understand the human by simply analyzing the individual as a stable configuration of traits, qualities, or attitudes," argues Robert Lauer, recalling Heidegger (*Temporal Man: The Meaning and Uses of Social Time* (New York: Praeger, 1981), 56), convincingly adding that exactly the same is true for social interactions and society as well (86ff).